Demand specification comparison

The original Main and Partial IV tables use the same 7% incumbent sample and UPC-unweighted representative price, without additional controls or month FE. The product-size tables below use a separately reported complete-case sample.

N = 540; 132 month clusters. Coefficients shown with month-clustered standard errors in parentheses.

Alpha is the signed price slope in +alpha * price (expected negative). Nicotine enters uninstrumented in every model.

National count includes all eligible brands nationwide. Geographic count is the distinct-brand union over states where the focal firm sells.

Tax means the geographic population-weighted tax. In full IV, tax and count jointly instrument both endogenous regressors.

Main

	M1	M2	M3	M4	M5	M6
Estimator	Full IV	Full IV	Full IV	Full IV	OLS	OLS
Firm FE	No	No	Yes	Yes	No	Yes
Count IV	National	Geographic	National	Geographic	--	--
Tax IV	Yes	Yes	Yes	Yes	No	No
alpha: price	-0.211 (0.117)	-0.202 (0.122)	-0.339 (0.220)	-0.332 (0.225)	-0.035 (0.007)	-0.046 (0.016)
beta_a: nicotine	0.007 (0.013)	-0.000 (0.016)	0.037 (0.010)	0.035 (0.011)	0.055 (0.003)	0.057 (0.005)
sigma: nesting	3.422 (0.665)	3.630 (0.769)	4.024 (1.087)	4.159 (1.162)	0.975 (0.057)	0.911 (0.052)
N	540	540	540	540	540	540
Price first-stage F	62.93	63.11	36.32	36.81	--	--
Share first-stage F	55.00	39.17	48.63	37.91	--	--
sigma in [0,1)	No	No	No	No	Yes	Yes

Partial IV

	D1	D2	D3	D4	D5	D6
Estimator	Share-only IV	Share-only IV	Share-only IV	Share-only IV	Price-only IV	Price-only IV
Firm FE	No	No	Yes	Yes	No	Yes
Count IV	National	Geographic	National	Geographic	--	--
Tax IV	No	No	No	No	Yes	Yes
alpha: price	-0.461 (0.642)	-1.349 (6.524)	-0.263 (0.103)	-0.292 (0.131)	-0.303 (0.056)	-0.490 (0.109)
beta_a: nicotine	-0.025 (0.117)	-0.193 (1.216)	0.038 (0.010)	0.035 (0.011)	0.082 (0.008)	0.089 (0.012)
sigma: nesting	5.699 (7.026)	15.538 (71.921)	3.536 (1.165)	3.883 (1.520)	1.345 (0.058)	1.130 (0.062)
N	540	540	540	540	540	540
Price first-stage F	--	--	--	--	62.57	37.91
Share first-stage F	0.46	0.04	8.06	5.45	--	--

sigma in [0,1)	No	No	No	No	No	No

Interpretation and assumptions

- Partial-IV models are diagnostics: share-only IV treats price as exogenous; price-only IV treats within-nest share as exogenous. Their first stages include that untreated regressor as a control.

- OLS treats both regressors as exogenous; an admissible OLS sigma does not validate that assumption.

- First-stage F entries are joint tests of the excluded instruments using month-clustered covariance. They are not conditional weak-identification tests for the two-endogenous-regressor models. The CSV also includes IID F and partial R2.

- Month clustering allows within-month dependence, not serial dependence across months. No model here tests or establishes exclusion. All IV specifications are exactly identified.

- Full-IV sigma estimates remain outside [0,1); estimates are neither constrained nor clipped. No model is automatically selected for cost recovery.

- Each model was checked against an independent explicit-dummy QR OLS/2SLS calculation and structural-residual reconstruction.

- Product size has been constructed as a reusable characteristic; none of the estimates above include it. Size and variety specifications appear below.

Product size and UPC variety: common complete-case sample

All sixteen models use N = 528 and 132 month clusters. Excluded firm-months are listed in excluded_firm_months.csv; unresolved BLU UPC 088796900752 is not filled automatically. No incomplete-UPC mean is substituted; the entire firm-month is excluded. Existing shares, prices, market totals and instruments are retained.

Size is the unweighted mean mL per cartridge/disposable over eligible device UPCs. Bottles and mixed types are excluded from that mean. Variety is log(n_upcs), counting all distinct eligible UPCs (positive sales and at least three stores), including bottles. Nicotine, size and variety are treated as exogenous controls; endogenous assortment choice remains a concern. Price and within-nest log share are jointly instrumented by geographic tax and the indicated count. No month FE. Standard errors are clustered by month.

Baseline

	B1	B2	B3	B4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.113 (0.065)	-0.103 (0.069)	-0.002 (0.070)	0.005 (0.071)
beta_a	0.013 (0.010)	0.007 (0.011)	0.036 (0.007)	0.035 (0.007)
beta_size	--	--	--	--
beta_variety	--	--	--	--
sigma	2.887 (0.356)	3.036 (0.408)	2.384 (0.231)	2.409 (0.234)
N	528	528	528	528
p_it first-stage F	74.35	74.60	35.30	35.78
ln_s_i_g first-stage F	73.52	51.21	84.50	70.46

Size

	S1	S2	S3	S4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.115 (0.081)	-0.100 (0.085)	-0.024 (0.075)	-0.015 (0.076)
beta_a	0.012 (0.009)	0.008 (0.010)	0.015 (0.009)	0.015 (0.009)
beta_size	-0.018 (0.146)	0.024 (0.155)	-0.214 (0.091)	-0.208 (0.091)
beta_variety	--	--	--	--
sigma	2.880 (0.340)	3.045 (0.400)	2.267 (0.190)	2.299 (0.195)
N	528	528	528	528
p_it first-stage F	68.21	68.72	34.95	35.28
ln_s_i_g first-stage F	53.86	36.58	70.61	67.22

Variety

	V1	V2	V3	V4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.189 (0.095)	-0.187 (0.100)	0.011 (0.065)	0.019 (0.066)
beta_a	0.006 (0.014)	0.001 (0.015)	0.040 (0.007)	0.039 (0.007)
beta_size	--	--	--	--
beta_variety	-0.483 (0.182)	-0.523 (0.202)	0.134 (0.103)	0.130 (0.106)
sigma	3.316 (0.563)	3.487 (0.641)	2.315 (0.236)	2.347 (0.241)
N	528	528	528	528
p_it first-stage F	84.32	82.61	36.67	36.94
ln_s_i_g first-stage F	52.10	40.79	67.65	57.27

Size and variety

	SV1	SV2	SV3	SV4
Firm FE	No	No	Yes	Yes
Count IV	National	Geographic	National	Geographic
alpha	-0.189 (0.095)	-0.187 (0.101)	-0.005 (0.067)	0.004 (0.068)
beta_a	0.006 (0.013)	0.001 (0.015)	0.022 (0.009)	0.021 (0.009)
beta_size	-0.033 (0.166)	0.003 (0.175)	-0.201 (0.084)	-0.194 (0.085)
beta_variety	-0.459 (0.185)	-0.525 (0.213)	0.173 (0.099)	0.167 (0.102)
sigma	3.281 (0.528)	3.491 (0.623)	2.186 (0.192)	2.227 (0.198)

N	528	528	528	528
p_it first-stage F	69.77	69.73	36.50	36.83
ln_s_i_g first-stage F	34.15	27.61	60.12	55.49

First-stage F values are joint month-clustered tests of excluded instruments, not conditional weak-identification statistics. Size exogeneity is an assumption. Reviewed retail pack-volume corrections are propagated through the rebuilt sales panel; the NCP market-size input is unchanged. All coefficients and structural residuals passed an independent explicit-dummy QR 2SLS check.

Compare S, V and SV against B with the same instrument and fixed effects to isolate control effects. All sixteen columns retain the same sample, including variety-only models that do not require size themselves.

Detailed models, coefficients, first stages, checks, estimation sample and excluded firm-months are saved under specification_comparison/product_size/.

Market-size sensitivity (NCP inputs unchanged)

All 48 specifications use the same 528 complete-size firm-month observations and 132 month clusters. The four baseline, size-only, variety-only and combined specifications are each estimated under three market-size constructions. See excluded_firm_months.csv for exclusions.

Annual: the within-year arithmetic mean of household-weighted monthly NCP purchasing intensity times that year's July adult population. Monthly: each month's household-weighted purchasing intensity times the same annual adult population. Constant: the full-period arithmetic mean of the monthly intensity series times annual adult population. No NCP input was rebuilt.

For each alternative, s_it = firm mL / M_t and s_0t = 1 - total market mL / M_t are recomputed. The within-nest share, prices, nicotine, controls, roster and instruments remain fixed. This isolates the market-size construction; coefficients use signed alpha and month-clustered standard errors (see CSVs). All outside shares were checked positive.

Model	Controls	Firm FE	Count IV	Sigma annual	Sigma monthly	Sigma constant	Alpha annual	Alpha monthly	Alpha constant
B1	Baseline	No	National	2.887	2.773	3.103	-0.113	-0.105	-0.302
B2	Baseline	No	Geographic	3.036	2.902	3.296	-0.103	-0.097	-0.289
B3	Baseline	Yes	National	2.384	2.255	2.952	-0.002	0.008	-0.294
B4	Baseline	Yes	Geographic	2.409	2.271	2.999	0.005	0.012	-0.281
S1	Size	No	National	2.880	2.771	2.916	-0.115	-0.106	-0.359
S2	Size	No	Geographic	3.045	2.914	3.144	-0.100	-0.093	-0.338
S3	Size	Yes	National	2.267	2.160	2.641	-0.024	-0.010	-0.351
S4	Size	Yes	Geographic	2.299	2.181	2.705	-0.015	-0.004	-0.334
V1	Variety	No	National	3.316	3.142	3.979	-0.189	-0.171	-0.458
V2	Variety	No	Geographic	3.487	3.291	4.192	-0.187	-0.169	-0.455
V3	Variety	Yes	National	2.315	2.169	3.054	0.011	0.025	-0.314
V4	Variety	Yes	Geographic	2.347	2.191	3.099	0.019	0.030	-0.303
SV1	Size and variety	No	National	3.281	3.123	3.469	-0.189	-0.171	-0.462
SV2	Size and variety	No	Geographic	3.491	3.304	3.759	-0.187	-0.169	-0.457
SV3	Size and variety	Yes	National	2.186	2.067	2.683	-0.005	0.012	-0.361
SV4	Size and variety	Yes	Geographic	2.227	2.096	2.748	0.004	0.019	-0.346

All 48 models passed independent QR 2SLS coefficient and structural-residual checks. Size and variety exogeneity remain maintained assumptions. Full coefficients, clustered standard errors, first stages and admissibility checks are saved in specification_comparison/market_size/.